Arista Access Node ELEC-ACC-01 QoS Congestion Event SOP (ALARM_3339)

Node: ELEC-ACC-01 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_3339

Description

This Standard Operating Procedure (SOP) is designed to address QoS congestion events, specifically microbursts and interface tail drops, on the Arista Access Node ELEC-ACC-01 at the Electronic City Hub. QoS congestion events occur due to the finite capacity of the node's buffers and queues, which can become overwhelmed by excessive traffic bursts. On Arista's EOS (Extreme Operating System), this is further exacerbated by the use of shared buffers and queues for multiple interfaces, leading to potential queue starvation and tail drop events. Furthermore, the lack of explicit policing and shaping mechanisms on the Access Node can contribute to the unchecked growth of bursty traffic, ultimately resulting in congestion events and ALARM_3339 triggers. This SOP aims to identify and mitigate the root causes of QoS congestion events on ELEC-ACC-01, ensuring optimal network performance and minimizing potential disruptions.

Pre-requisite Checks

- Check: Verify the Electronic City Hub is accessible and all necessary personnel have been dispatched.

- Check: Confirm the redundant link status and ensure it is functioning as expected.

- Check: Review the event logs for any recent configuration changes or software upgrades.

Potential Root Causes

Cause	Probability
Uncontrolled microbursts on interfaces due to lack of policing	High
Queue starvation due to uneven traffic distribution	Medium
Insufficient buffer allocation for interfaces	Low

Diagnostic Commands & Expected Outputs

```
Command: show hardware queue, Expected Output: Queue statistics and buffer usage for all interfaces., Description: This command provides detailed information about the queue usage and buffer allocation for each interface, helping to identify potential areas of congestion and queue starvation.

Command: show qos police, Expected Output: Summary of all policing configurations on the node., Description: This command reveals the current policing configurations in place on the Access Node, helping to determine if any interfaces are experiencing excessive traffic bursts or if there are any gaps in policing mechanisms.
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```
Command: show interface statistics, Expected Output: Detailed statistics for each interface, including errors, drops, and packets transmitted., Description: This command provides detailed interface statistics, including errors, drops, and packets transmitted, which can help identify the source of the congestion event and potential areas for intervention.
```

Resolution Steps

Step 1: Step: Implement interface policing for the affected interfaces, Command: configure qos police interface bandwidth conform-action drop exceed-action drop, Description: This step involves configuring policing mechanisms on the affected interfaces to prevent excessive traffic bursts and ensure that traffic conforms to the specified bandwidth limits.

Step 2: Step: Adjust buffer allocation for the affected interfaces, Command: configure hardware queue buffer-size , Description: This step involves adjusting the buffer allocation for the affected interfaces to ensure that there is sufficient buffer space to handle the expected traffic loads.

Step 3: Step: Configure class-based policing for even traffic distribution, Command: configure qos class-map match any, Description: This step involves configuring a class map to match all traffic and then policing that traffic to ensure even distribution and prevent queue starvation.

Post-Check Verification

```
Command: show hardware queue, Expected Output: Queue statistics and buffer usage for all interfaces should be within expected ranges., Description: This command is used to verify that the queue usage and buffer allocation for each interface are within expected ranges after implementing the resolution steps.

Command: show interface statistics, Expected Output: Detailed statistics for each interface should show reduced errors, drops, and improved packet transmission., Description: This command is used to verify that the interface statistics have improved after implementing the resolution steps, indicating that the congestion event has been mitigated.
```

Roll-back Procedures

- Step: Remove policing configurations for the affected interfaces, Command: no configure qos police interface , Description: If the fix fails, this step involves removing the policing configurations for the affected interfaces to restore the original configuration.

- Step: Restore original buffer allocation for the affected interfaces, Command: no configure hardware queue , Description: If the fix fails, this step involves restoring the original buffer allocation for the affected interfaces to its previous settings.